

DAY 5 REPORT

Resume Text Extraction Engine

Objective

Implemented a resume text extraction engine that extracts raw text from PDF and DOCX resumes for use in an ATS pipeline.

Deliverables

- Resume text extraction engine
- Cleaned resume outputs
- Test result logs

Implementation Summary

- Built parsers/resume_text_extractor.py
- Added PDF extraction using pdfplumber
- Added DOCX extraction using python-docx
- Automatically detects file extension
- Successfully tested using multiple resumes

Day 5 – Resume Text Extraction

Since Day 5 involved actual coding, use snippets from your implementation.

PDF Text Extraction

```
def extract_pdf_text(file_path):  
    text = ""  
  
    with pdfplumber.open(file_path) as pdf:  
        for page in pdf.pages:  
            page_text = page.extract_text()  
  
            if page_text:  
                text += page_text + "\n"  
  
    return text
```

DOCX Text Extraction

```
def extract_docx_text(file_path):
    doc = Document(file_path)

    text = ""

    for paragraph in doc.paragraphs:
        text += paragraph.text + "\n"

    return text
```

Automatic File Detection

```
def extract_resume_text(file_path):
    file_path = Path(file_path)

    if file_path.suffix.lower() == ".pdf":
        return extract_pdf_text(file_path)

    elif file_path.suffix.lower() == ".docx":
        return extract_docx_text(file_path)

    else:
        raise ValueError("Unsupported file format")
```

Execution Result

The screenshot displays the VS Code interface with the following components:

- EXPLORER:** Shows the project structure with folders like 'data', 'resumes', and 'docs'. The file 'main.py' is selected.
- EDITOR:** Displays the code for 'main.py', which imports 'extract_resume_text' and calls it on a sample resume file.
- TERMINAL:** Shows the command 'python main.py' being executed. The output indicates that the file 'data/resumes/ai-developer-resume.docx' was selected and its text was successfully extracted.

The terminal output is as follows:

```
PS E:\Zecser Technology\Tasks Completed\zecpath-ai-system> python main.py
Selected file: data/resumes/ai-developer-resume.docx
Extension: .docx

Reading DOCX: data/resumes/ai-developer-resume.docx
Paragraphs found: 34

=====
Sophia Martinez
Northwood, OH 43623
(555)555-5555 - Sophia.Martinez@example.com
Professional Summary
Innovative AI Software Developer with expertise in machine learning, utilizing Python and TensorFlow to deliver impactful solutions. Proven track record in boosting predictive power and optimizing neural networks.
Work History
AI Software Developer
January 2024 to August 2025
DeepTech Innovations - Northwood, OH
Developed AI algorithms, boosting accuracy by 20%
Implemented a new ML pipeline, reducing processing time
Optimized neural networks, improving efficiency by 35%
Machine Learning Engineer
January 2022 to December 2023
Quantum AI Systems - Cleveland, OH
Designed ML models, increasing predictive power by 15%
```

Console output confirming successful extraction across sample resumes

The screenshot shows a VS Code editor with a project named 'zecpath-ai-system'. The file explorer on the left shows a directory structure with folders like 'data', 'parsers', and 'utils'. The main editor displays a file named 'main.py' with the following Python code:

```

1 from parsers.resume_text_extractor import extract_resume_text
2
3
4 def main():
5
6     file_path = "data/resumes/data-scientist-resume.pdf"
7
8     text = extract_resume_text(file_path)
9
10    print("\n=====")
11    print(text)
12    print("=====")
13
14
15 if __name__ == "__main__":
16     main()

```

The terminal at the bottom shows the command 'python main.py' being executed, resulting in the following output:

```

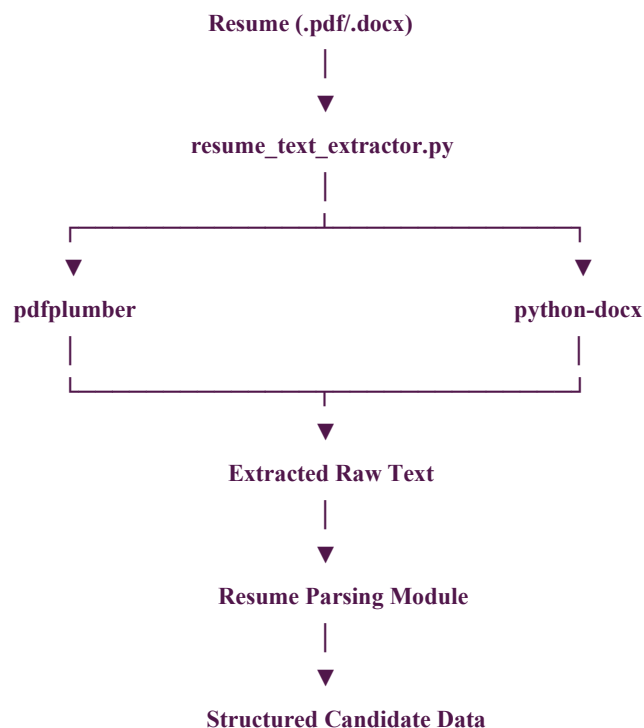
PS E:\Zecser Tecnology\Tasks Completed\zecpath-ai-system> python main.py
===== Extracted Resume Text =====
GRACE Entry-Level Data Scientist | Python & Machine Learning
JACKSON +1-(234)-555-1234 Email linkedin.com Indianapolis, Indiana
VOLUNTEERING SUMMARY
Data Analytics Volunteer Driven to leverage a strong foundation in data science to develop impactful
analytical models for retail challenges. Proficient in Python, R, SQL, and
Food Rescue US
experienced in creating data-driven solutions. Aspires to advance analytical
01/2025 - 01/2025 expertise while contributing to innovative projects in the retail analytics sector.
• Increased reporting efficiency by
enhancing data tracking systems. EXPERIENCE
• Optimized logistics data to
streamline delivery routes. Data Science Intern 01/2026 - 01/2026
Clarity Analytics Remote
EDUCATION
• Assisted in the cleansing and preprocessing of over 200,000 data records,
Bachelor of Science in Data improving dataset quality for analysis.
• Developed a predictive sales model using Python, increasing forecast accuracy
by 15% for quarterly sales meetings.
Purdue University • Collaborated with marketing teams to design data-driven strategies, enhancing
campaign effectiveness by analyzing customer behavior.
01/2023 - 01/2026 West Lafayette, IN

```

Console output confirming successful extraction across sample resumes

Resume Extraction Pipeline

This is perfect for your Day 5 report.



Resume Text Extraction Engine

Objective: Build a parser to extract raw text from PDF and DOCX resumes.

Architecture

Resume (.pdf/.docx) → Text Extractor → PDF/DOCX Parser → Extracted Text → Resume Parser

Deliverable	Status
PDF Extraction	Completed
DOCX Extraction	Completed
Extension Detection	Completed
Testing	Completed

Expected Deliverables

- Create automated test runs

Conclusion

The extraction engine successfully reads PDF and DOCX resumes and produces clean text output. This completes Day 5 and prepares the project for structured resume parsing.

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